

SNEZHNOGORSK, Russian Federation

WMO: 231790

Lat: **68.100N**Lon: **87.767E**Elev: **308**StdP: **14.53**Time Zone: **7.00 (E07)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.3	-44.1	-53.3	0.2	-48.9	-48.7	0.3	-44.2	17.8	-2.9	15.7	-1.3	4.6	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	79.1	64.7	74.4	62.3	70.5	60.4	66.1	76.1	63.9	72.3	61.6	68.4	4.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.4	85.2	69.5	60.4	79.3	67.6	58.2	73.1	66.3	31.0	76.4	29.4	72.6	27.7	68.0	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.1	13.1	-51.4	84.8	4.5	3.9	-54.6	87.6	-57.3	89.9	-59.8	92.1	-63.1	95.0		
(5)			-51.4	67.6	4.5	3.3	-54.6	70.0	-57.3	72.0	-59.8	73.8	-63.1	76.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	-17.3	-11.9	3.1	16.2	29.2	50.4	61.1	52.8	42.0	22.1	-5.7	-8.0
	DBStd	29.16	17.11	15.16	15.98	11.23	8.18	9.53	8.04	7.01	7.77	11.10	14.29	15.66
	HDD50	11698	2085	1733	1453	1014	646	113	7	49	262	865	1672	1799
	HDD65	16608	2550	2153	1918	1464	1111	447	175	383	691	1330	2122	2264
	CDD50	631	0	0	0	0	0	124	351	135	21	0	0	0
	CDD65	66	0	0	0	0	0	8	54	4	0	0	0	0
	CDH74	420	0	0	0	0	0	57	360	3	0	0	0	0
	CDH80	76	0	0	0	0	0	7	69	0	0	0	0	0
Wind	WSAvg	6.4	6.4	6.6	7.6	7.7	7.0	6.1	4.8	5.3	5.8	7.0	5.7	6.3
Precipitation	PrecAvg	33.7	2.8	2.4	2.4	2.3	2.4	2.4	2.5	3.2	2.9	4.3	3.1	3.2
	PrecMax	44.5	5.4	4.3	6.2	4.8	3.9	4.3	4.7	5.6	5.9	6.9	5.6	5.8
	PrecMin	20.9	0.6	0.6	0.4	0.2	0.8	0.6	0.2	1.1	1.3	1.3	0.6	0.4
	PrecStd	4.9	1.3	1.0	1.2	1.1	0.8	0.9	1.2	1.0	1.0	1.4	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	22.8	33.8	41.9	52.9	80.3	85.7	73.5	64.0	47.4	30.2	24.6
		MCWB	18.2	21.5	31.6	33.7	42.5	61.3	67.2	63.1	55.2	43.9	29.8	23.8
	2%	DB	13.0	18.7	31.3	36.9	46.6	74.1	81.9	70.4	59.3	40.8	26.0	21.3
		MCWB	12.0	17.6	28.7	32.3	38.5	59.8	66.1	61.3	52.6	38.3	25.4	20.4
	5%	DB	8.9	13.2	28.8	34.3	42.6	69.2	78.3	66.9	56.0	37.4	22.2	17.8
		MCWB	8.0	12.0	26.5	31.2	36.5	57.4	64.8	60.1	51.4	35.6	21.5	16.9
	10%	DB	4.8	8.6	25.8	32.2	39.4	64.8	74.3	63.3	52.2	34.8	15.1	13.8
		MCWB	3.9	7.5	23.6	29.7	34.7	55.5	62.7	58.0	48.1	33.0	14.2	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	22.1	32.0	35.4	43.9	65.9	70.0	64.7	56.5	44.1	29.7	23.9
		MCDB	19.1	22.8	33.7	38.4	50.9	75.6	81.0	70.2	61.6	47.2	29.9	24.4
	2%	WB	12.1	17.5	29.1	33.4	39.5	61.4	67.7	63.0	54.3	39.1	25.5	20.5
		MCDB	12.9	18.5	31.2	36.2	45.1	71.2	78.8	68.6	58.2	40.1	25.8	21.2
	5%	WB	8.1	12.0	26.6	31.7	37.2	58.5	65.7	61.2	51.8	36.0	21.4	17.0
		MCDB	8.9	13.0	29.0	34.1	41.4	67.9	76.0	65.8	55.7	37.3	22.3	17.9
	10%	WB	3.8	7.5	23.8	29.4	35.3	56.1	63.9	58.8	48.8	33.3	14.3	12.9
		MCDB	4.8	8.5	25.7	32.1	38.6	63.7	72.9	62.5	51.9	34.6	15.1	13.9
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.1	13.6	14.7	11.6	13.8	14.4	11.0	8.2	8.0	10.5	12.0
		MCDBR	11.8	13.6	8.7	14.4	14.1	20.3	18.4	15.5	10.7	8.7	10.0	13.0
		MCWBR	11.3	12.4	8.2	11.9	8.9	11.2	9.2	7.9	6.6	7.6	10.5	12.6
	5% WB	MCDBR	11.7	12.7	8.4	13.2	12.3	18.3	15.9	11.9	10.1	8.4	10.1	13.0
		MCWBR	11.2	12.4	8.0	11.7	9.0	11.7	9.0	6.9	7.0	7.7	10.4	13.0
Clear-Sky Solar Irradiance	taub	0.215	0.256	0.307	0.330	0.360	0.375	0.409	0.395	0.331	0.257	0.199	N/A	N/A
	taud	1.851	1.973	1.998	2.048	2.112	2.349	2.270	2.270	2.389	2.100	1.881	N/A	N/A
	Ebn at Noon	20	160	220	254	260	261	245	233	223	160	20	0	0
	Edh at Noon	3	17	30	38	40	33	35	31	21	14	3	0	0
All-Sky Solar Radiation	RadAvg	9	155	638	1312	1773	1720	1603	991	536	183	29	0	0
	RadStd	2	17	69	113	144	153	211	112	58	35	4	0	0